**Time-frequency representations of ERPs and multivariate statistical analysis**

TUTORIAL IIC

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## Synopsis

In the last part of the tutorial, you will see the Complex Wavelet Transform (CWT) being applied to the EEG dataset you already worked with in part A. Since the data is very high dimensional, the computation of the CWT time-frequency results both takes a while (~20 min) and generates a very large data file, also (~ 8GB). As this is unpractical for this tutorial, you will work with the pre-computed time-frequency data. The following steps were taken for the CWT computation – the information below should allow you to replicate the results:

1. Complex Wavelet Transform
   - Family of complex Morlet wavelets of kernel length: 2s
   - Frequencies: 2-40 Hz in steps of 2 Hz
   - Number of cycles: 3-10 (linearly increasing from lowest to highest frequency)
   - Computation of CWT for each frequency, channel, trial, and subject individually.
2. Computation of the CWT magnitude
3. Single-trial baseline correction: subtraction of frequency-specific magnitude within period -300 to -100 ms for each trial, channel, and subject individually
4. Downsampling time-dimension by factor 4 (to reduce file-size)

The objective of this tutorial is to explore the time-frequency results and apply different statistical analyses to the data. The goal is to quantitatively work out the main effects in time, frequency and spatial dimensions present in the data.

## Dataset structure

The TF-DATA file is a MATLAB struct similar to the one used in part A of the tutorial.

| | |
|---|---|
| ***.magnitude*** | 5D-matrix containing the magnitudes of the CW transformed EEG data (27 channels * 20 frequencies * 128 timepoints * 150 trials * 17 subjects) |
| ***.data_dimensions*** | A cell array containing information about the dimensions in ***.magnitude*** |

| | |
|---|---|
| ***.chanlocs*** | Contains both the channel labels and channel location in two different coordinate systems (needed for topographic visualization) |
| ***.times*** | A vector associating sample-count with time-information (-500 to 1500 ms) |
| ***.frex*** | A vector containing the frequencies of the wavelet family |
| ***.subjects*** | A cell-array containing the subject IDs |
| ***.srate*** | The sampling rate |
| ***.labelcodes*** | A cell-array containing the meaning of the numerical labels in ***.label*** |
| ***.labels*** | A 2D-array containing the labels of all trials and subjects |
| ***.unit*** | The physical unit of the data (µV – microvolts) |
| ***.chanlist*** | A list of channel-labels (complementary to ***.chanlocs***) |

## 1: Computation of the grand average time-frequency difference; exploration of the results using tfviewerx

Download the file TF_DATA.mat (see download link) and load it into the MATLAB workspace. Explore the file structure as described above and ask questions if anything is unclear.

Compute the grand average difference time-frequency magnitude. For this, you first need to compute the per-subject average time-frequency magnitude for each condition separately and then subtract the condition-wise grand averages (error minus non-error).

Use `tfviewerx` (in the folder 'functions') to visualize the resulting time-frequency representation. Change some of the setting interactively, such as color limits, time-range, etc. Finally, identify the strongest effects in terms of the maximal and minimal activations. Discuss the findings with your classmates. What might these effects originate from? In which way do they relate to the time-domain ERP investigated in part A of this tutorial?

## 2: Univariate statistical testing without correction for multiple comparisons

Next, you will perform individual (independent) level-2 statistical tests for each of the variables (bins/pixels: channels * frequencies * time-points) in the dataset.

Compute the per-subject difference average and subsequently test each variable separately against the null hypothesis using the ordinary t-test (`ttest`) with a p-value threshold of 0.05. Create a *stats-mask*: a Boolean matrix of size (channels * frequencies * time-points) containing ones for statistically significant bins and zeros for statistically insignificant bins.

Visualize the grand average difference time-frequency representation using `tfviewerx`, but this time by applying the *stats-mask* to the data (element wise multiplication of the stats-mask with the grand average difference matrix). This way only the statistically significant bins will be visualized.

Explore the visualization and look for statistically significant effects. What can you observe?

## 3: Univariate statistical testing with Bonferroni correction

A famous and widely used method for correcting statistical analyses for multiple comparisons is the Bonferroni correction. The basic principle is the reduction of the p-value threshold by a factor equal to the number of comparisons (or number of tested variables).

Compute the adjusted p-value and recompute the stats-mask with the new threshold. Hint: use the p-values computed in section 1. Visualize the grand average difference time-frequency representation again using `tfviewerx`, this time with the Bonferroni corrected *stats-mask*.

What can you observe in comparison with the uncorrected results?

## 4: Non-parametric permutation testing with correction for multiple comparisons

A suitable method for statistical analysis of the data we are dealing with is non-parametric permutation testing. Its basic principle is to create random data surrogates and compare the distributions of these surrogates with the values in the original data. If these values lie outside of a pre-defined percentile of the surrogate distributions, we assume these to be significant.

Follow below steps to apply the method to the dataset:

1. Compute the grand average difference time-frequency results by following the procedure in section 1.
2. Shuffle the original labels for each subject individually and compute a surrogate (based on the shuffled labels) of the grand average difference time-frequency results. Hint: use the MATLAB function `randperm` to create random permutations of vectors.
3. Find the global minimum and maximum activation (across all channels, frequencies and time-points) of the surrogate and store both values separately.
4. Repeat steps 2 and 3 for at least *nPerm=250* times. This creates a distribution of global minimum and maximum values of size *nPerm.* Remark: Depending on your implementation, this may take a few minutes to be computed.

5. Visualize the minimum and maximum distributions in two separate histograms and compute the 5% confidence thresholds: determine the threshold for values smaller than the 2.5% percentile of the minimum distribution (*min-threshold*) and the threshold for values greater than the 97.5% percentile of the maximum values (*max-threshold*). Hint: use the function *prctile* to compute the thresholds.
6. Compare the original values of step 1 with the percentile thresholds obtained in the previous step. Any bin containing values smaller than the *min-threshold* and greater than the *max-threshold* are denoted significant.

Create a *stats-mask* as in sections 2 and 3. Visualize the grand average difference time-frequency representation again using `tfviewerx`, this time with the non-parametric permutation-based *stats-mask*.

What can you observe in comparison with the uncorrected and Bonferroni corrected results? Which of the effects identified in section 1 finally turned out statistically significant, which not?

References / further reading:

[1] Cohen, M. X. (2014). *Analyzing neural time series data: theory and practice*. MIT press.

[2] http://mikexcohen.com/lectures.html